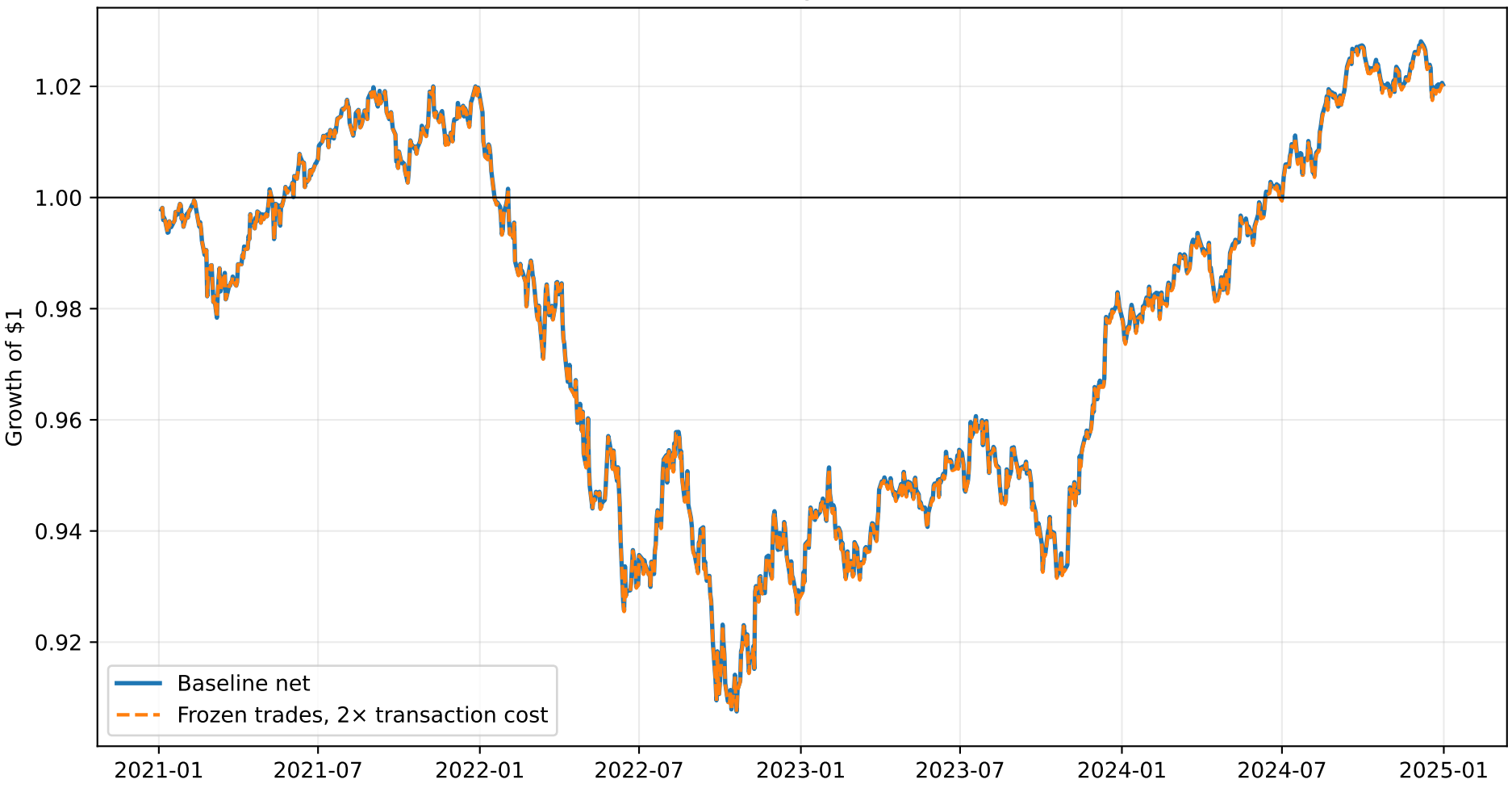
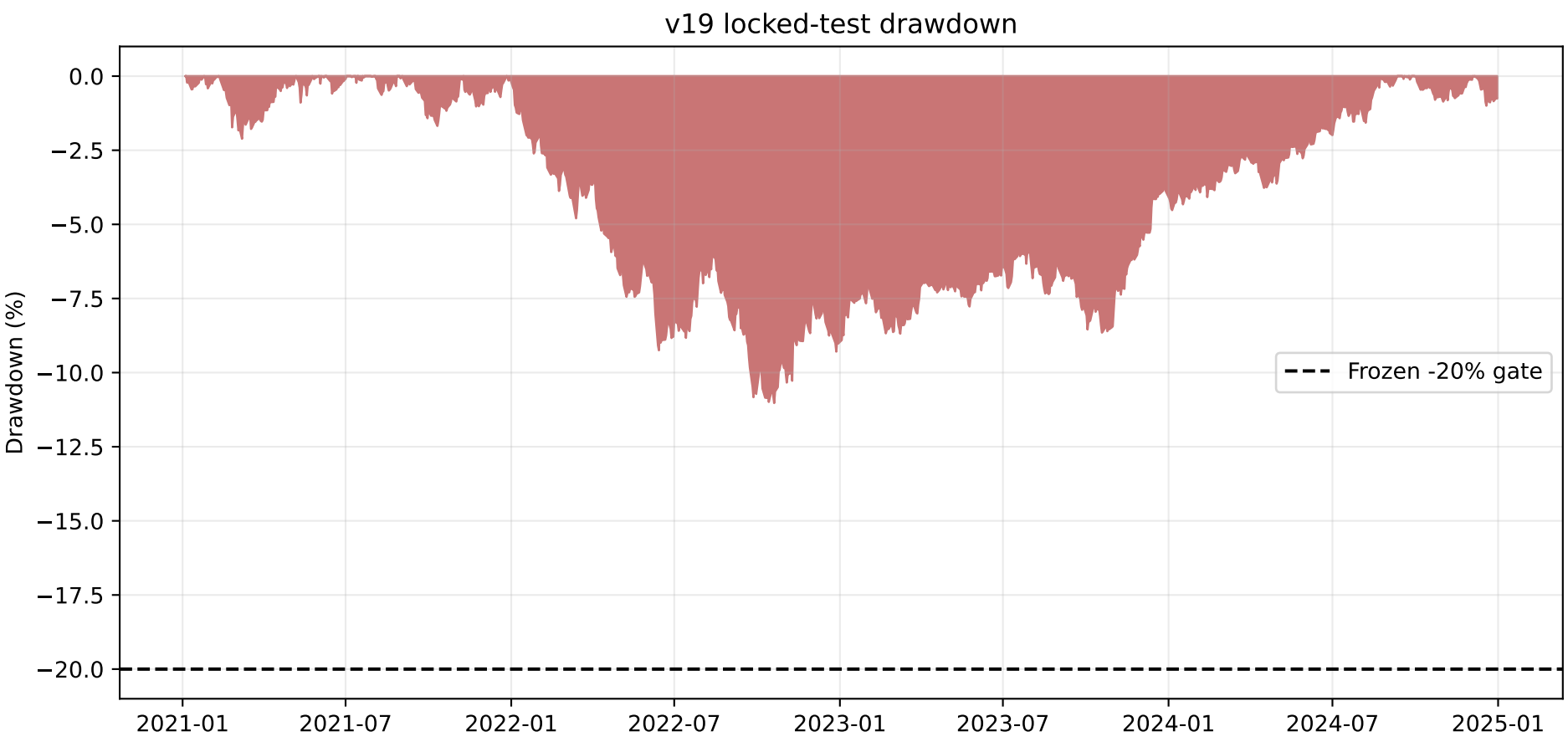
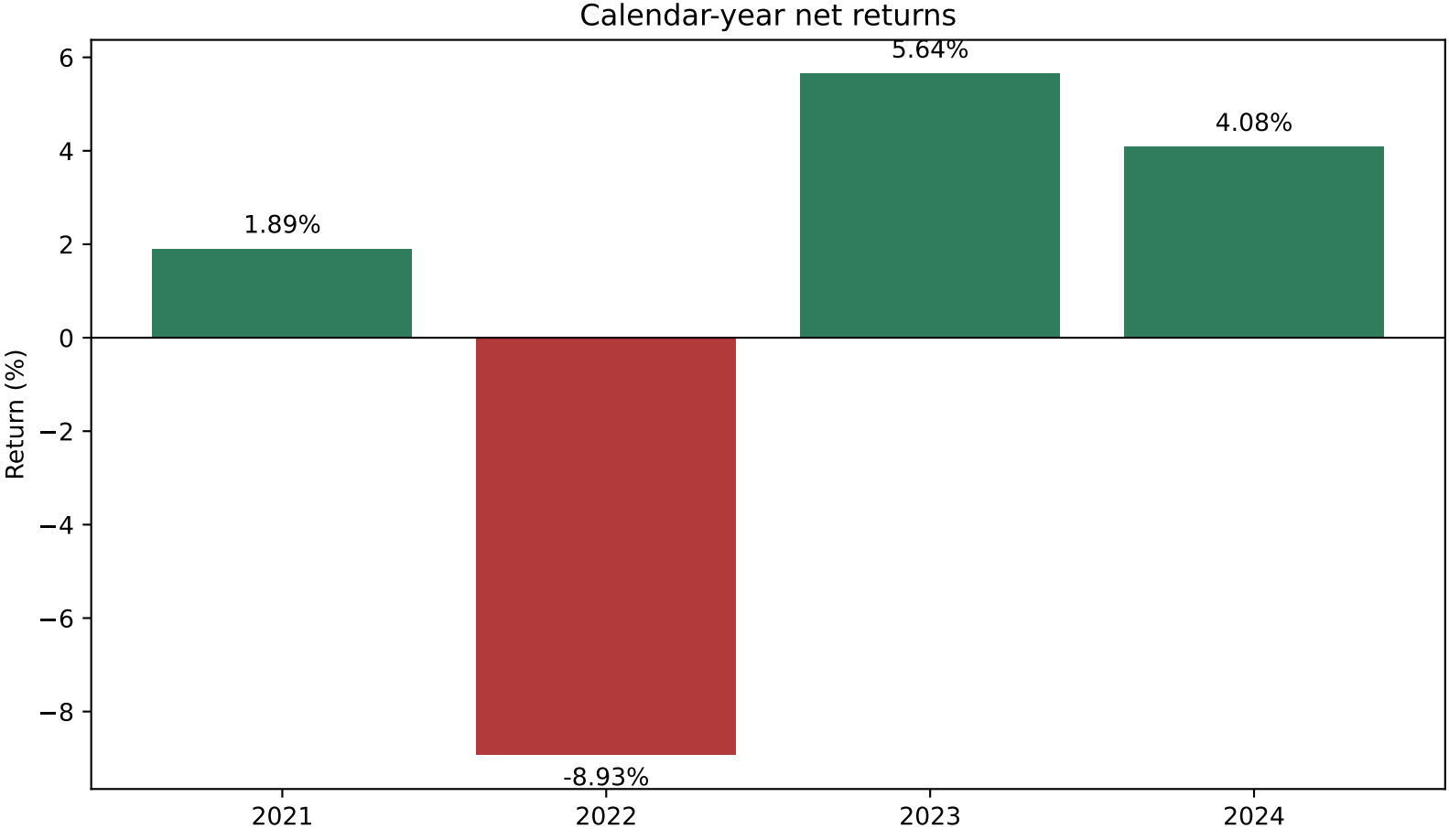
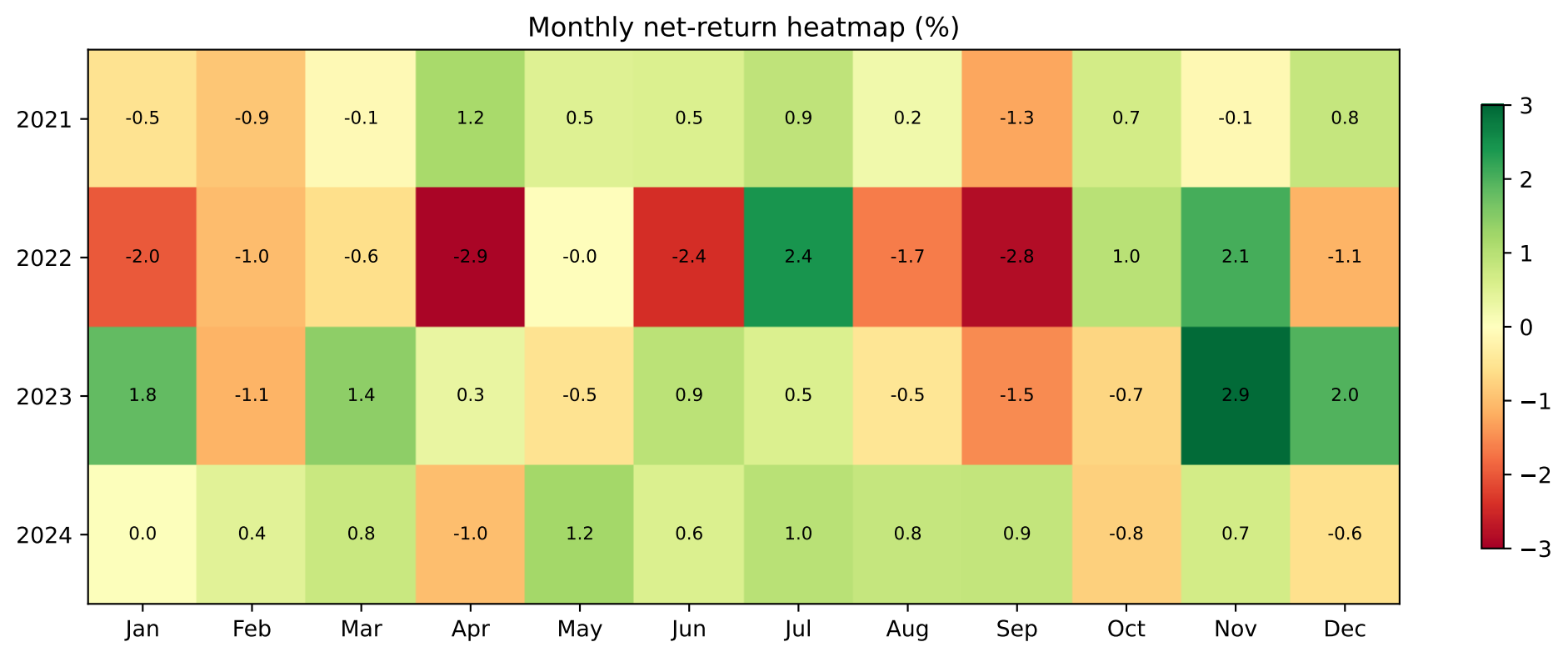
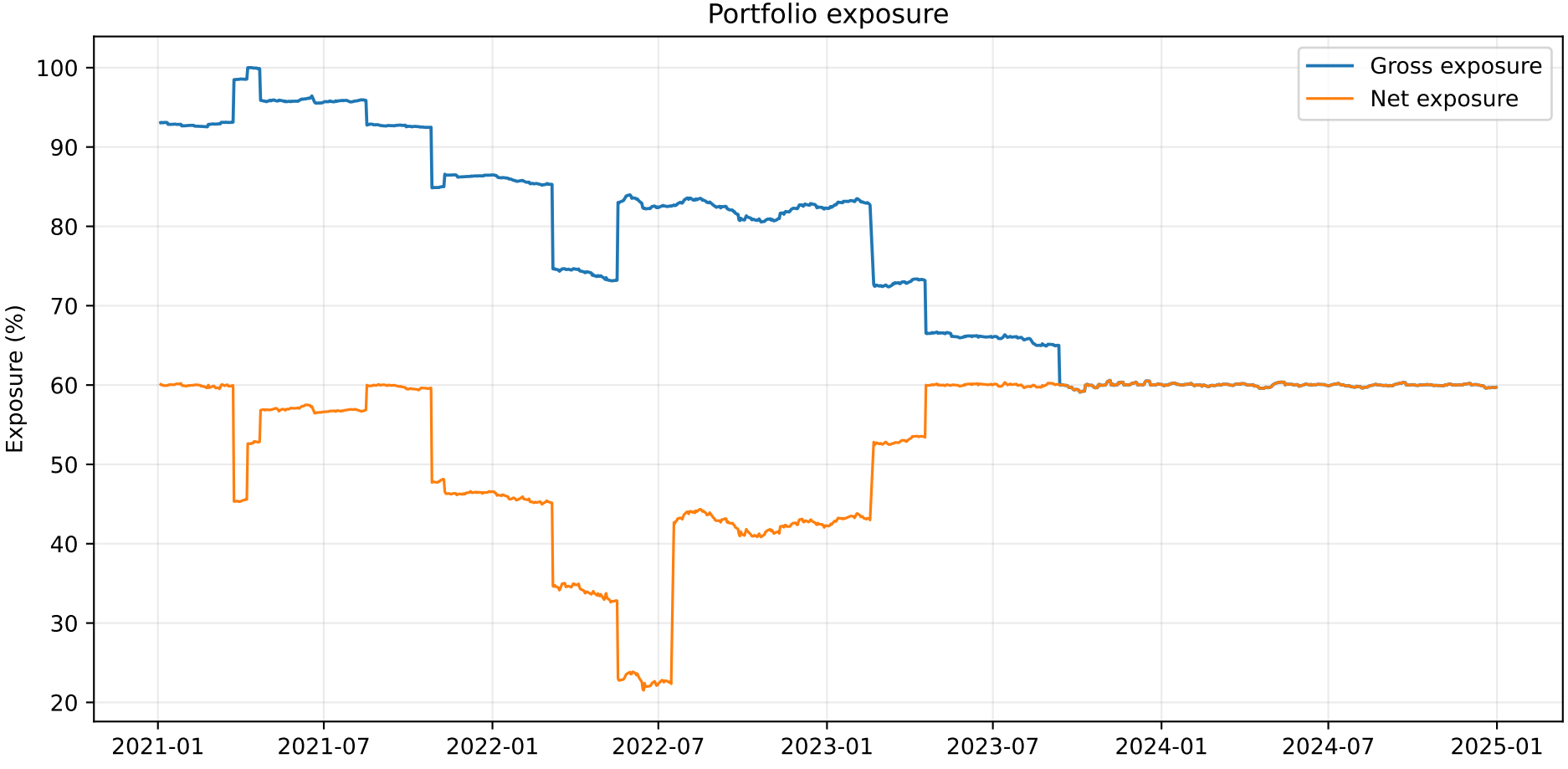
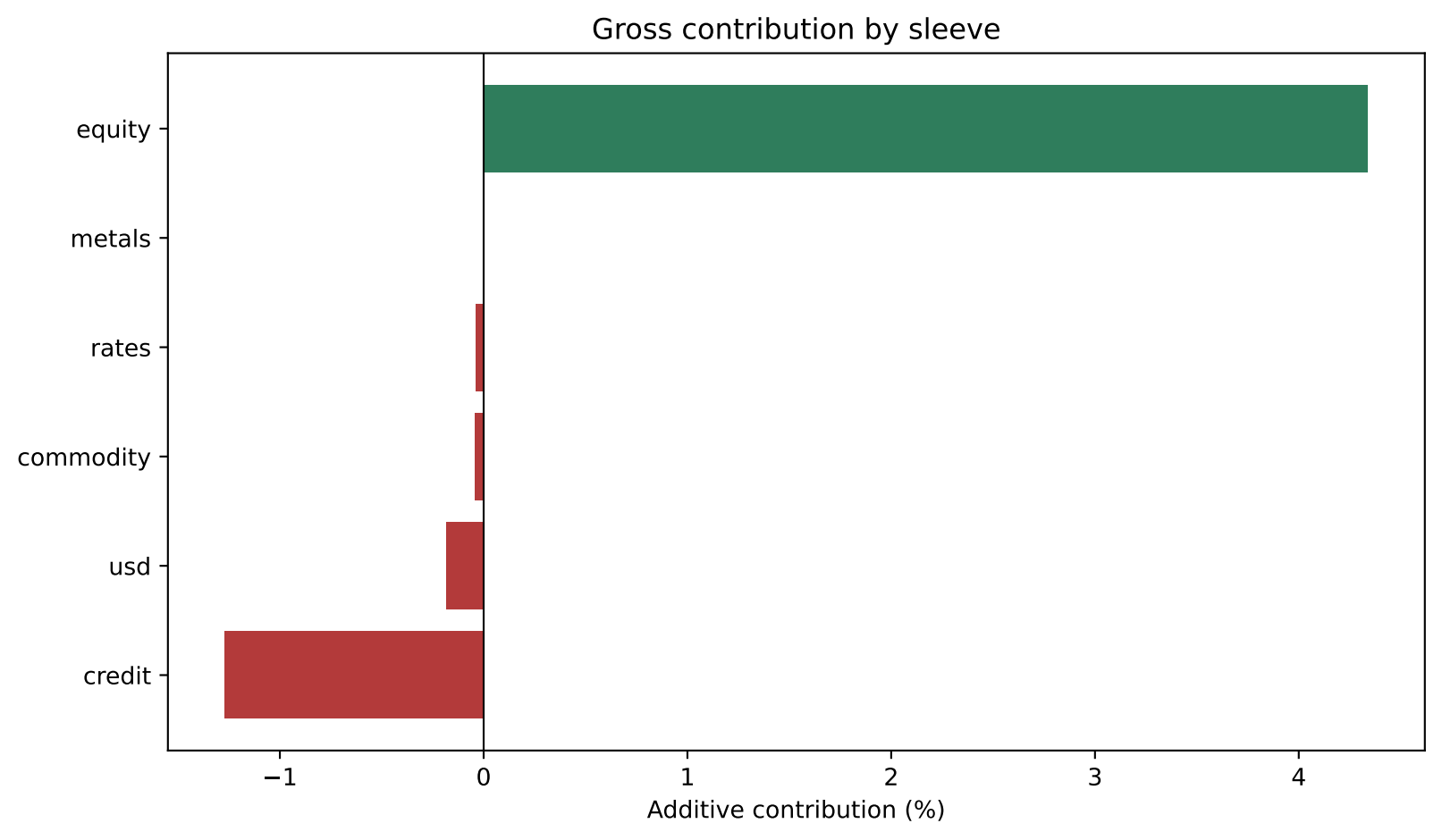

v19 locked-test net growth (2021–2024)
Growth of $1
1.02
1.00
0.98
0.96
0.94
0.92
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
Baseline net
Frozen trades, 2× transaction cost

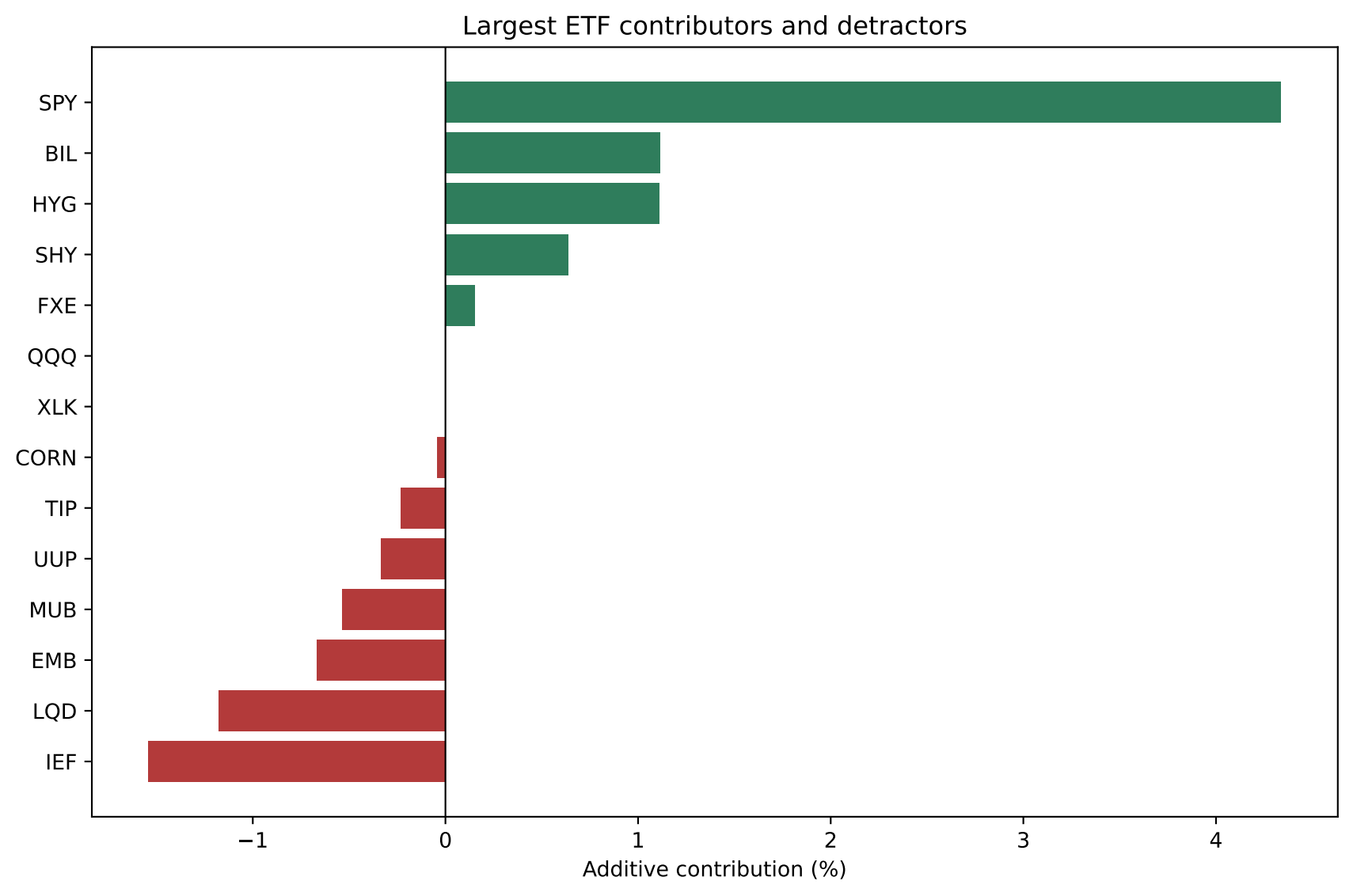

v19 locked-test drawdown
Drawdown (%)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
Frozen -20% gate

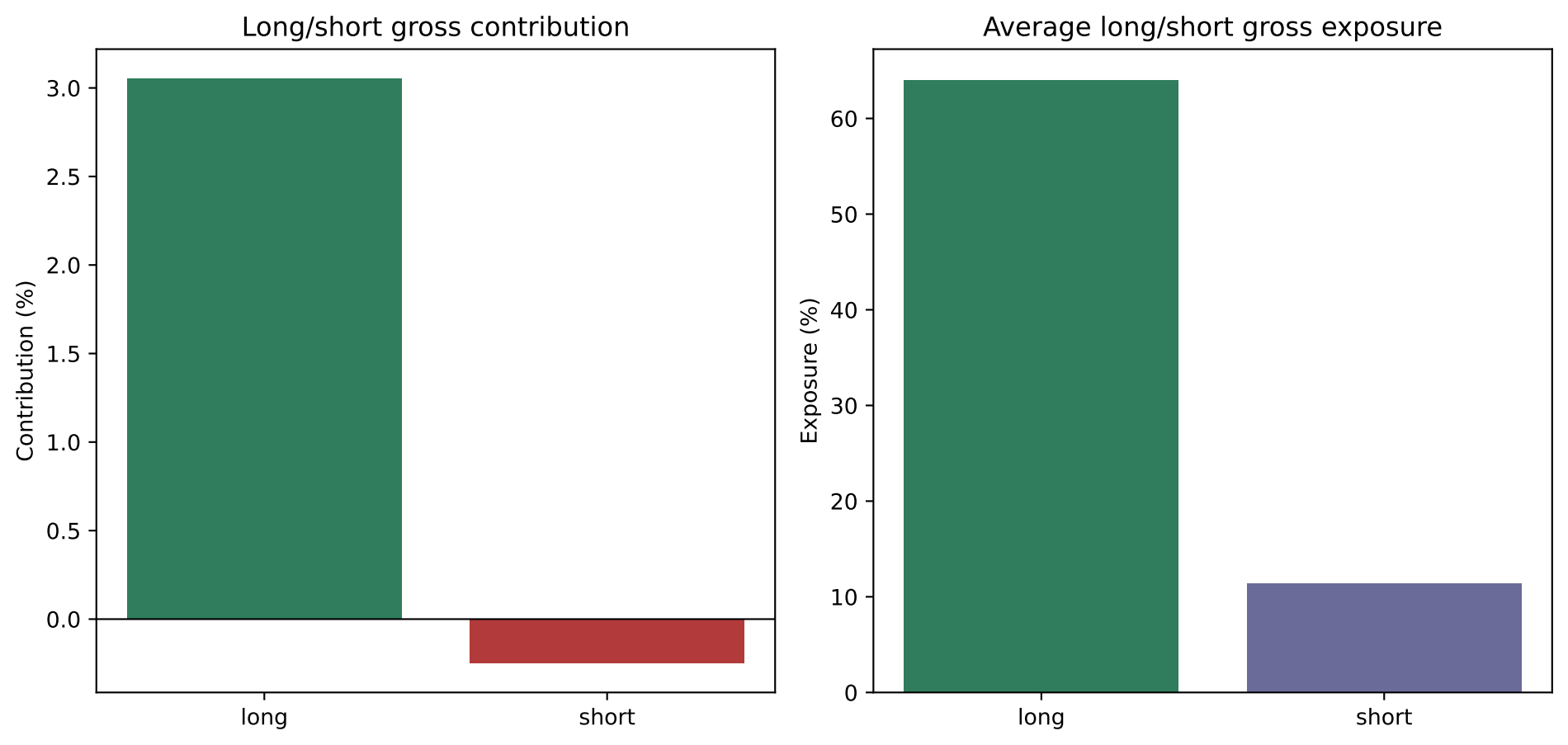

Calendar-year net returns
Return (%)
6
4
2
0
−2
−4
−6
−8
1.89%
-8.93%
5.64%
4.08%
2021
2022
2023
2024

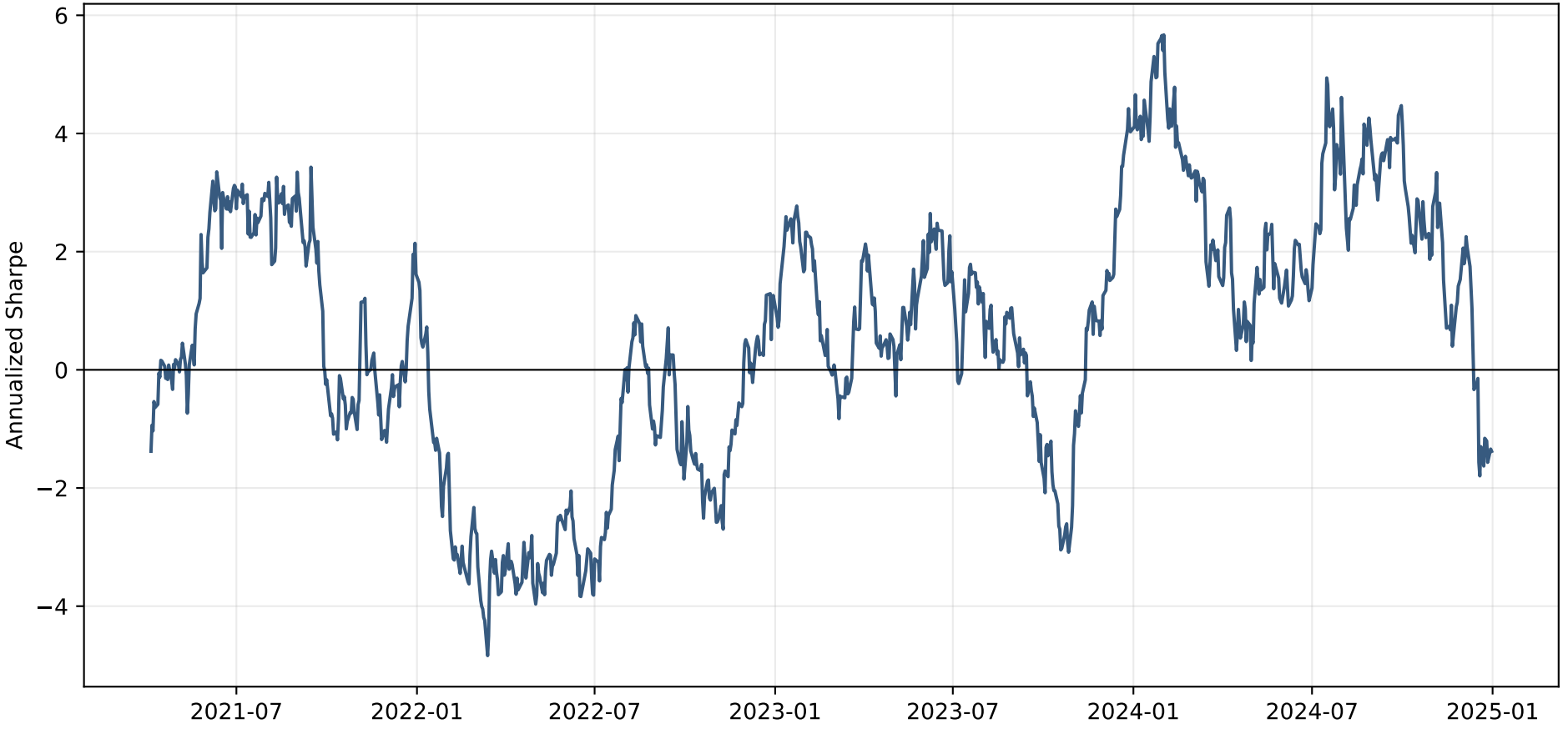

Monthly net-return heatmap (%)
2021
2022
2023
2024
-0.5 -0.9 -0.1 1.2 0.5 0.5 0.9 0.2 -1.3 0.7 -0.1 0.8
-2.0 -1.0 -0.6 -2.9 -0.0 -2.4 2.4 -1.7 -2.8 1.0 2.1 -1.1
1.8 -1.1 1.4 0.3 -0.5 0.9 0.5 -0.5 -1.5 -0.7 2.9 2.0
0.0 0.4 0.8 -1.0 1.2 0.6 1.0 0.8 0.9 -0.8 0.7 -0.6
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
3
2
1
0
−1
−2
−3

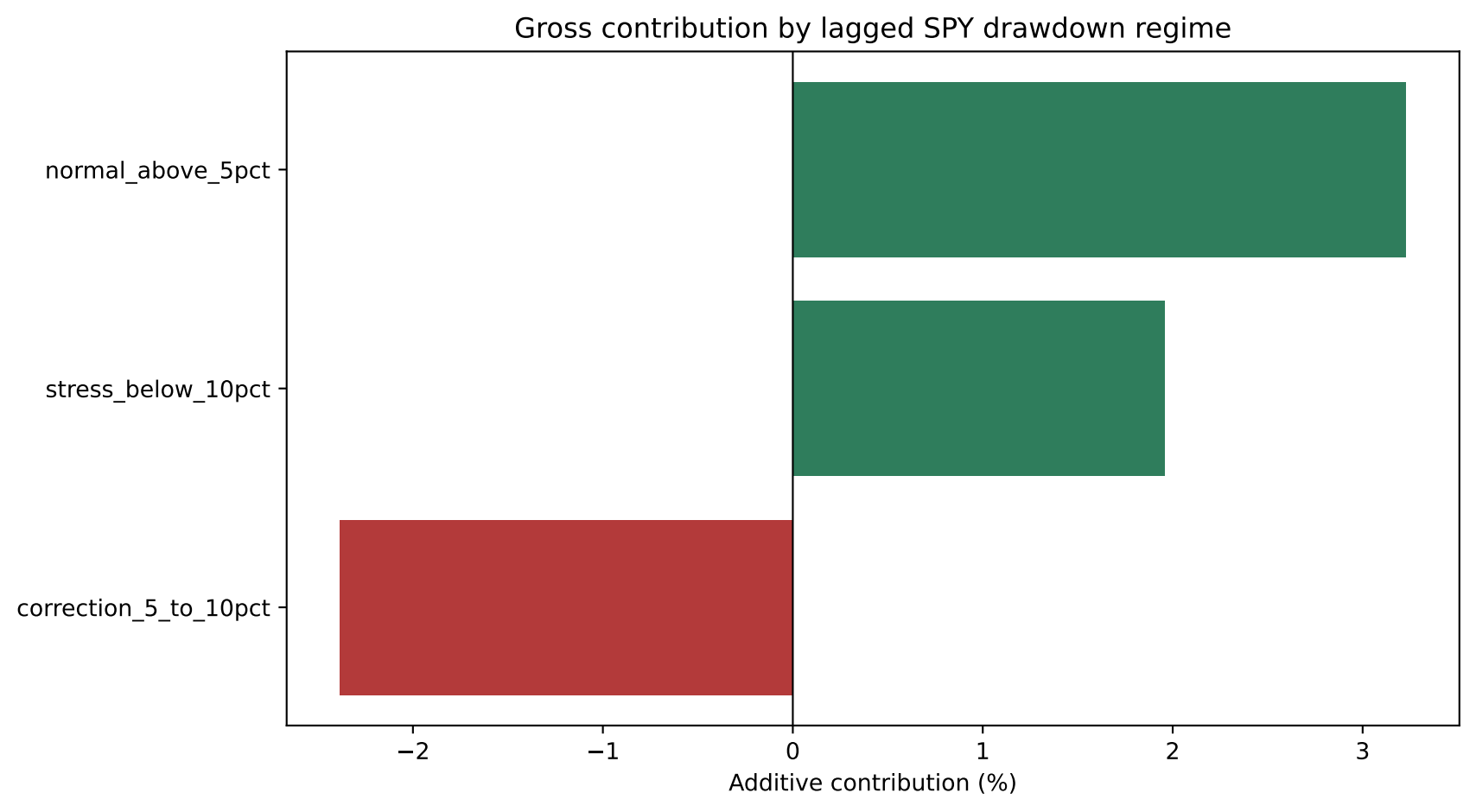

Portfolio exposure
Gross exposure
Net exposure
Exposure (%)
100
90
80
70
60
50
40
30
20
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01


Gross contribution by sleeve
equity
metals
rates
commodity
usd
credit
−1
0
1
2
3
4
Additive contribution (%)


Largest ETF contributors and detractors
SPY
BIL
HYG
SHY
FXE
QQQ
XLK
CORN
TIP
UUP
MUB
EMB
LQD
IEF
−1
0
1
2
3
4
Additive contribution (%)


Long/short gross contribution
Contribution (%)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
long
short
Average long/short gross exposure
Exposure (%)
60
50
40
30
20
10
0
long
short


63-session rolling net Sharpe
Annualized Sharpe
6
4
2
0
−2
−4
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01


Gross contribution by lagged SPY drawdown regime
normal_above_5pct
stress_below_10pct
correction_5_to_10pct
−2
−1
0
1
2
3
Additive contribution (%)